SHAREMATE: A SOCIALLY CONNECTED SMART MOBILITY AND COMMUNITY SUPPORT APP

INDIVIDUAL THESIS

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Individual component:

Public Transport Live Sharing for University Students – FriendZone LiveTransit Share Engine (FLTS Engine)

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DECLARATION AND SUPERVISOR CERTIFICATION

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Abstract

University students who depend on public buses frequently experience uncertainty about vehicle location, arrival time, congestion, and service quality, especially where buses are not equipped with dedicated tracking hardware. This individual research develops the FriendZone LiveTransit Share Engine (FLTS Engine), the public-transport intelligence component of the ShareMate mobile application. The component uses consent-based passenger smartphone GPS observations to infer bus movement, forms route-based FriendZones so students on the same route receive relevant information, estimates arrival times collaboratively, supports map-based traffic and incident reporting, and links identified bus licence plates with community passenger feedback. The research follows a design-oriented development approach supported by literature on crowdsourced transport sensing, bus arrival-time prediction, trust and reputation, vehicle identification, and microservice architectures. The solution is designed as a microservice-oriented mobile system using Flutter for the client, Python/FastAPI services, Firebase for authentication and real-time data management, and map services for route visualisation. Preliminary results reported for the ShareMate research indicate 94.2% GPS/location processing accuracy and 91.4% ETA prediction accuracy for the FLTS component, while the complete ShareMate system reported a 420 ms average response time. The principal individual contribution is a decentralized, passenger-powered public-transport sharing model focused on university commuting, reducing dependence on bus-installed GPS infrastructure while combining live tracking, FriendZones, ETA, incidents, bus identification, and reputation in one workflow. The component provides a practical foundation for scalable community-based transport intelligence and future large-scale evaluation with real university commuters.
Keywords: crowdsourced public transport, FriendZone, GPS, ETA prediction, university mobility

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CNN	Convolutional Neural Network
ETA	Estimated Time of Arrival
FLTS	FriendZone LiveTransit Share Engine
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ITS	Intelligent Transport System
OCR	Optical Character Recognition
REST	Representational State Transfer
SLIIT	Sri Lanka Institute of Information Technology
UAT	User Acceptance Testing
UI	User Interface
UX	User Experience

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1 INTRODUCTION

1.1 Overall project context

ShareMate is a university-focused digital platform designed to support collaborative transportation, academic resource sharing, private ride sharing, and community-based grocery assistance within a single environment. The project addresses the fragmentation of student services, where transport, learning resources, and everyday support are often managed through separate applications or informal communication channels. ShareMate brings these services together through independent components that communicate via APIs, allowing each service to be developed and scaled separately [17], [18].

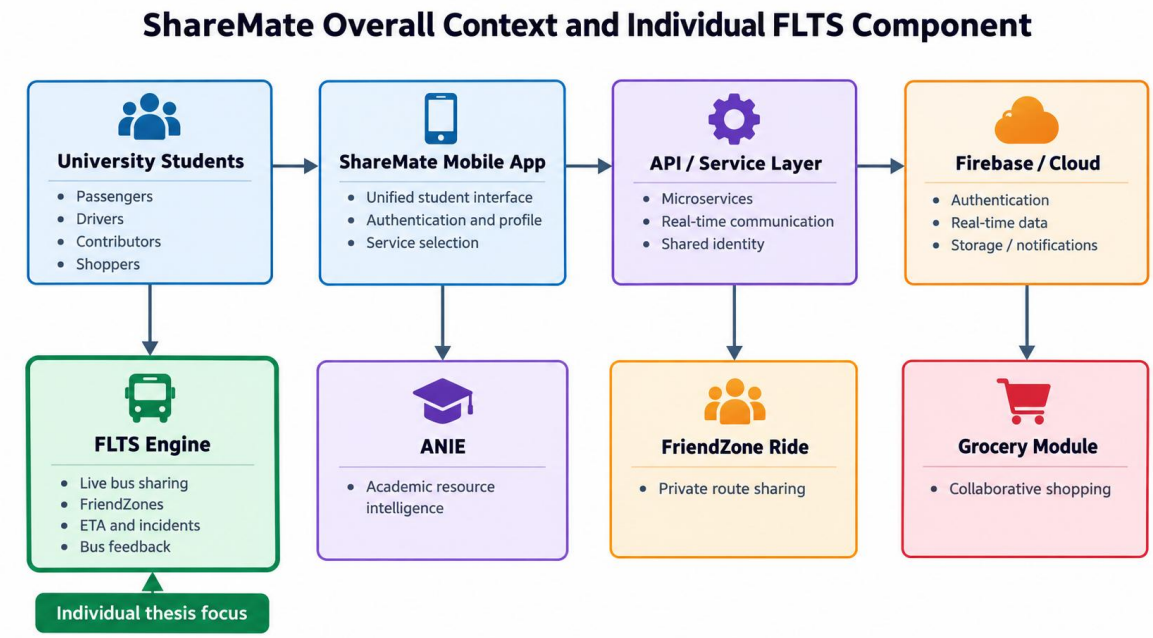


Figure 1.1 ShareMate overall context and individual FLTS component

1.2 Individual component context

This thesis focuses on the FriendZone LiveTransit Share Engine (FLTS Engine), the public-transport component of ShareMate. FLTS is intended for university students who travel by bus and need timely information about bus position, expected arrival time, route disruptions, and the service history of a particular vehicle. Rather

than relying only on transport operators to generate real-time data, the component allows passengers to act as distributed sensing participants by contributing mobility observations through their smartphones.

The FLTS concept combines six closely related functions: user registration and identity management, passenger-powered live bus location sharing, route-based FriendZones, collaborative ETA estimation, traffic and incident reporting, and bus identification with passenger feedback. These functions are intended to operate as a coherent workflow rather than as isolated features.

1.3 Problem statement

Many university students rely on public buses but cannot reliably determine where a bus is, when it will arrive, whether the route is congested, or what previous passengers experienced on that vehicle. Conventional real-time tracking normally assumes that the vehicle itself is equipped with dedicated GPS hardware and that a central transport authority publishes the data. Where such infrastructure is absent or incomplete, passengers still experience uncertainty and long waiting times.

A second problem is the fragmentation of information. Students may know about a delay, accident, or road blockage, but such information is often shared informally through messaging groups and is not automatically associated with the relevant bus route. Similarly, feedback about a particular bus is rarely connected to a simple vehicle-identification process that future passengers can consult. The research problem is therefore to determine how a low-cost, student-centered public-transport intelligence mechanism can use passenger-generated data while maintaining acceptable accuracy, timeliness, privacy, and usability.

1.4 Research gap

Previous research provides strong foundations for bus arrival-time prediction, crowdsourced incident reporting, shared mobility, trust and reputation systems, licence plate recognition, and scalable microservice architectures [3], [4], [5], [7]–[12], [17]–[19]. However, these capabilities are generally studied as separate problems. The identified gap is the absence of a university-focused transport-sharing component that combines passenger-generated live bus location, route communities,

collaborative ETA estimation, route-specific incident reporting, and bus-level reputation within one mobile workflow without depending on dedicated vehicle-tracking hardware.

1.5 Aim and objectives

The main aim is to design, implement, and evaluate a community-powered public-transport live-sharing component for university students that uses passenger smartphone data to improve visibility of bus movement and route conditions.

Table 1.1 Individual FLTS objectives and success indicators

ID	Specific objective	Success indicator
O1	Enable authenticated university users to share consent-based GPS observations during bus journeys.	Successful secure sharing session and valid timestamped observations.
O2	Form route-based FriendZones that distribute relevant live transport information to students on the same route.	Correct route grouping and relevant update delivery.
O3	Estimate bus position and collaborative ETA using multiple passenger observations and route progress.	Position/ETA accuracy within defined evaluation targets.
O4	Allow students to pin congestion, accidents, road blockages, and other incidents to a route.	Incident stored, mapped, and delivered to relevant FriendZone.
O5	Identify buses by licence plate and associate them with passenger feedback and service history.	Correct record lookup and successful feedback retrieval/submission.
O6	Evaluate the component for accuracy, latency, usability, privacy, security, and integration with ShareMate.	Measured technical results and user/stakeholder validation.

1.6 Research questions or hypotheses

- RQ1: Can passenger-generated smartphone GPS observations provide sufficiently reliable information to infer bus movement without dedicated bus tracking hardware?

- RQ2: Does route-based grouping make live transit information more relevant and manageable for university commuters?
- RQ3: Can collaborative movement observations support practical ETA estimation for upcoming route points?
- RQ4: Can community incident reporting and bus feedback improve situational awareness and service transparency while preserving user privacy?

1.7 Scope limitations and assumptions

The first implementation phase targets university students and bus routes commonly used by those students. The system assumes that participating passengers possess smartphones with GPS and mobile data, grant location permission during an active sharing session, and select or are associated with the correct route. The component does not replace an official transport authority, guarantee exact timetable adherence, or provide safety-critical navigation. ETA quality depends on the availability and freshness of passenger observations and route data. Bus feedback represents community experience and should not be interpreted as an official operator assessment.

1.8 Individual contribution

The individual contribution is the design of the FLTS Engine as a passenger-powered transport intelligence framework. The work brings together live passenger GPS sharing, automatic route-based FriendZones, collaborative ETA estimation, map-based incident pinning, and bus identification with feedback. The distinctive contribution is not any one of these mechanisms in isolation, but their integration around university commuting and their operation without requiring dedicated GPS hardware on the bus.

1.9 Thesis structure

Chapter 2 reviews the literature and related systems. Chapter 3 describes the research methodology and evaluation strategy. Chapter 4 presents product and solution design, including the FLTS architecture and detailed workflows. Chapter 5 describes implementation, integration, and testing. Chapter 6 presents preliminary

results and validation. Chapter 7 discusses the findings, contributions, and limitations. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the thesis and identifies future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This review focuses on the research areas directly relevant to the FLTS Engine: public-transport arrival prediction, crowdsourced mobility sensing and incident reporting, shared mobility and social influence, reputation mechanisms, vehicle identification, and microservice-based software architecture. The ShareMate research paper provides the primary source set for this thesis, and the studies cited there are used to position the individual component. Accordingly, the review explains how established techniques inform FLTS rather than attempting a broad survey of all intelligent transport systems.

2.2 Domain and theoretical foundations

FLTS lies within intelligent transportation systems, mobile crowdsensing, and community-based digital services. In mobile crowdsensing, ordinary smartphones become distributed sensors that contribute location or context observations. For public transport, multiple passenger observations can be interpreted collectively to infer the movement of a shared vehicle. The approach is especially attractive where installing and maintaining dedicated tracking equipment across the bus fleet is difficult.

Arrival-time prediction is a second foundation. Previous work has explored real-time redesign based on arrival predictions, Seq2Seq recurrent neural networks, deep-learning approaches, and Kalman-filter based frameworks [7]–[9], [19]. These studies demonstrate that ETA can be treated as a dynamic prediction problem rather than a static timetable lookup. FLTS adapts this principle to passenger-generated observations.

Trust and reputation are also important in a community-generated information system. Studies of the sharing economy highlight the importance of trust, service quality, legitimacy, and process transparency [4]–[6]. FLTS applies this idea through passenger feedback linked to identified buses and through reliability concepts for contributed location observations.

2.3 Related research

Dynamic and shared mobility research demonstrates the value of route matching, scheduling, and real-time decision support [1], [2]. Research on university shared travel also indicates that commuting behavior and service factors influence student adoption [3]. These findings support the decision to focus the initial deployment on university commuters, whose repeated travel routes create a natural basis for route-oriented communities.

Crowdsourced traffic research has shown that user-generated incident information can provide timely transportation awareness [10]. FLTS incorporates this principle as a route-linked "Pin the Traffic" mechanism where users can mark congestion, accidents, blockages, or other disruptions directly on the map.

Vehicle identification research demonstrates that license plates can be detected and recognized using computer-vision techniques such as deeply learned convolutional neural networks [12]. FLTS uses the identified plate as a stable key for retrieving passenger feedback and creating a community bus service history.

2.4 Existing systems products and techniques

Existing public-transport applications typically rely on official vehicle feeds, timetables, or GPS hardware installed on the vehicle. Community traffic applications demonstrate the power of user reporting, while navigation systems offer strong map and routing functions. However, these services are not necessarily designed around route-based university communities, nor do they combine passenger-generated bus tracking, collaborative ETA, incidents, and bus-specific reputation in one student-centered workflow.

Table 2.1 Synthesis of related work and FLTS response

Area	Existing contribution	Gap addressed by FLTS
Bus ETA prediction research [7]–[9], [19]	ETA models using historical/real-time data	Supports dynamic prediction, but not specifically university passenger-generated FriendZones.
Crowdsourced traffic reporting [10]	Passenger/user incident information	Provides timely incident data, but does not by itself infer a shared bus from passenger movement.

Area	Existing contribution	Gap addressed by FLTS
Shared mobility research [1]–[3], [11]	Route matching and adoption factors	Supports route community logic, but focuses mainly on ridesharing rather than live public-bus sharing.
Trust/reputation research [4]–[6]	Ratings and platform trust	Supports service-quality transparency, but not bus identification tied to crowdsourced tracking.
License plate recognition [12]	Vehicle identification using computer vision	Provides identification technique, but not integrated with route communities and passenger feedback.
FLTS Engine	All above functions in a university-focused public transport component	Combines passenger GPS, FriendZones, ETA, incidents, bus identification, and feedback without dedicated bus GPS hardware.

2.5 Gap synthesis and conceptual framework

The literature synthesis places the main novelty at the intersection of three ideas: passenger-powered sensing, route-based social grouping, and transport intelligence. Passenger-powered sensing supplies observations, FriendZones organize those observations and their recipients by route, and the transport-intelligence layer converts raw data into useful outputs such as bus position, ETA, route incidents, and bus reputation. This conceptual combination informs the methodology and architecture presented in the following chapters.

3 RESEARCH METHODOLOGY

3.1 Research design

This study adopts a design-oriented research approach in which a practical software artefact is developed to address a clearly defined commuting problem and is then evaluated using technical and user-centered measures. The methodology covers problem analysis, requirements definition, architecture design, incremental implementation, integration, testing, and validation. Development is iterative, allowing findings from technical tests and user feedback to guide subsequent refinements.

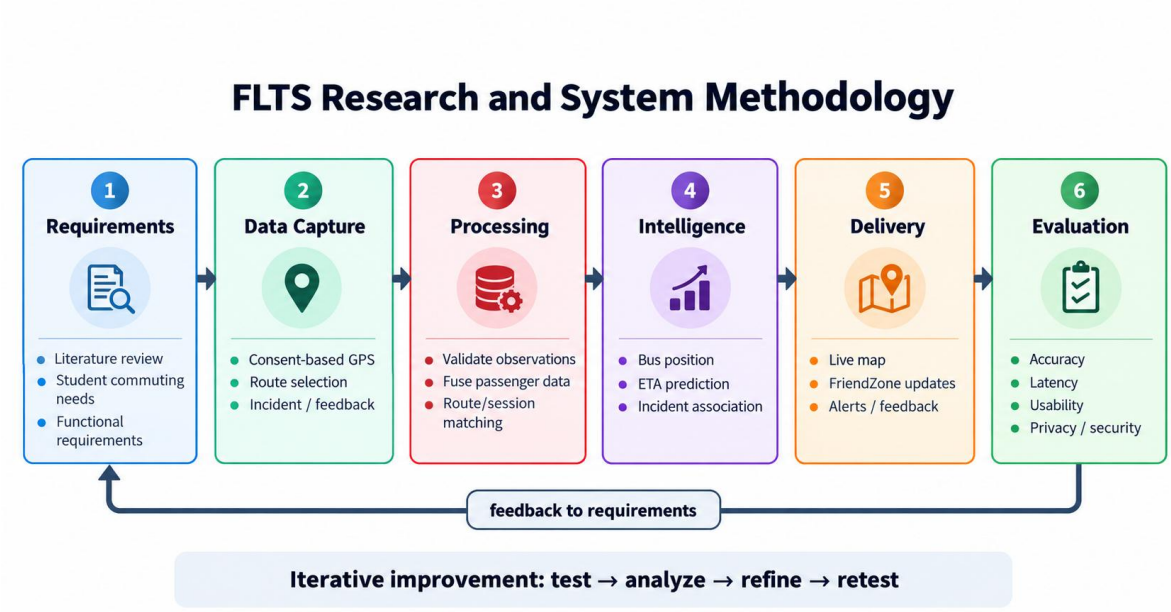


Figure 3.1 FLTS research and system methodology

3.2 Data participants or study materials

The principal operational data consist of user profiles, selected bus routes, consent-based smartphone GPS observations, timestamps, estimated speed, incident reports, bus identifiers, and passenger feedback. Each live observation can be represented conceptually as Li = (lati, longi, ti, vi), where latitude and longitude represent geographic position, ti is the observation time, and vi is the estimated

movement speed. Multiple passenger observations on the same route are combined to infer the shared bus movement.

The intended participant population is university students who regularly use public buses. For final human-participant evaluation, recruitment should follow institutional ethics requirements and use informed consent. The current ShareMate research reports component-level performance, but does not provide a detailed participant count for FLTS; therefore, this thesis does not invent one.

3.3 Experimental or study procedure

- Define selected university bus routes and route points for testing.
- Authenticate participating users and obtain explicit permission before location sharing.
- Start a live bus-sharing session when a participant is traveling on a selected route.
- Collect timestamped GPS observations at controlled intervals and associate them with the route and session.
- Validate observation freshness and route proximity, then combine multiple observations to estimate bus position.
- Calculate ETA for selected route points using current position, recent speed, route distance, and known incidents.
- Allow route participants to submit traffic incidents and associate each incident with the relevant route segment.
- Identify a bus using a licence plate value, retrieve previous passenger feedback, and allow new feedback after travel.
- Record accuracy, latency, errors, and user feedback; refine the system and repeat tests.

3.4 Measures and evaluation metrics

Table 3.1 Proposed evaluation metrics

Metric	Purpose	Measure
GPS/location processing accuracy	Correctness of inferred/processed position relative to accepted reference observations.	Percentage accuracy / positional error
ETA prediction accuracy	Agreement between predicted and observed arrival times.	Accuracy %, MAE in minutes where available
Response time	Time from request/update to system response.	Milliseconds
FriendZone correctness	Whether users and updates are associated with the correct route community.	Correct grouping rate
Incident delivery	Time and correctness of route-linked incident propagation.	Latency and route match rate
Plate identification	Correctness of bus plate extraction or matching.	Recognition accuracy
Usability	Ease of learning and completing core tasks.	UAT feedback / Likert score
Privacy/security	Correct permission, authentication, and access-control behavior.	Pass/fail security tests

3.5 Data analysis

Quantitative analysis focuses on accuracy and latency measures. ETA predictions should be compared with observed arrival times at selected route points, while the processed bus position should be compared with a trusted reference position from the same trip. Where sufficient observations are available, error distributions should be summarized using the mean, median, and range, and system latency should be reported using average and percentile response times. Qualitative UAT feedback should be organized into themes such as usefulness, clarity, trust, privacy concerns, and ease of use.

3.6 Ethics privacy security and responsible AI

Location information is sensitive because continuous coordinates can reveal travel patterns. FLTS therefore requires explicit opt-in before a live-sharing session, limits sharing to the duration necessary for the service, and should avoid exposing a user's precise identity to other passengers. Authentication and authorization must protect profile and route data, while backend rules must prevent unauthorized modification of incidents and feedback. Retention periods should be defined so that raw location data are not stored indefinitely without a research or service need.

Where computer vision is used for license plate recognition, the image should be processed only for vehicle identification and should not be repurposed for identifying people. Any AI-assisted ETA or recognition model should be evaluated for error, documented, and treated as decision support rather than an infallible source. Incident and feedback functions should include moderation and reporting controls to reduce misinformation, harassment, or reputational misuse.

3.7 Reproducibility and research artefacts

Reproducibility will be supported through version-controlled source code, documented API contracts, database schemas, test cases, configuration examples, and experiment notes. Sensitive credentials and private data must not be committed to the repository. Test datasets should be anonymized or synthetically reproduced where possible, and the exact software versions used for the final evaluation should be recorded in the repository and appendix.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The FLTS requirements translate the research problem into observable system behavior, with emphasis on real-time information exchange, route relevance, privacy, and integration with the wider ShareMate platform.

Table 4.1 Functional requirements

ID	Functional requirement
FR1	The system shall allow a student to register, authenticate, and maintain a university-linked profile.
FR2	The system shall allow an authenticated passenger to start and stop a live bus-location sharing session.
FR3	The system shall collect timestamped smartphone GPS observations with user consent.
FR4	The system shall associate observations with a selected bus route and travel direction.
FR5	The system shall create or join a route-based FriendZone and distribute relevant bus updates to its members.
FR6	The system shall estimate current bus position from one or more passenger observations.
FR7	The system shall calculate and display ETA for selected route points or stops.
FR8	The system shall allow users to pin traffic jams, accidents, road blockages, or delays on the map.
FR9	The system shall display active route incidents to relevant FriendZone users.
FR10	The system shall identify a bus using a licence plate number or recognized plate image.
FR11	The system shall retrieve historical passenger feedback for an identified bus.
FR12	The system shall allow a passenger to submit a rating and travel experience after using the bus.

Table 4.2 Non-functional requirements

ID	Non-functional requirement
NFR1 Usability	Core actions should be understandable to students with ordinary smartphone experience.
NFR2 Performance	Live location and incident updates should be delivered with low interactive latency.
NFR3 Reliability	Temporary network loss should not corrupt live sessions or route records.
NFR4 Scalability	Services should support growth in users, routes, and simultaneous live sessions.
NFR5 Security	Authentication, transport encryption, access control, and secure credential handling are required.
NFR6 Privacy	Precise location sharing must be opt-in, purpose-limited, and minimized.
NFR7 Maintainability	Microservices and APIs should allow independent component maintenance.
NFR8 Compatibility	The mobile application should support common Android/iOS devices supported by Flutter.

4.2 Component architecture within the complete solution

FLTS operates as an individual service domain inside ShareMate. The Flutter mobile client provides the user interface, Firebase supports identity and real-time application data, and Python/FastAPI services implement transport-specific logic. API-based separation allows FLTS to interact with shared authentication and notification services while remaining independently maintainable.

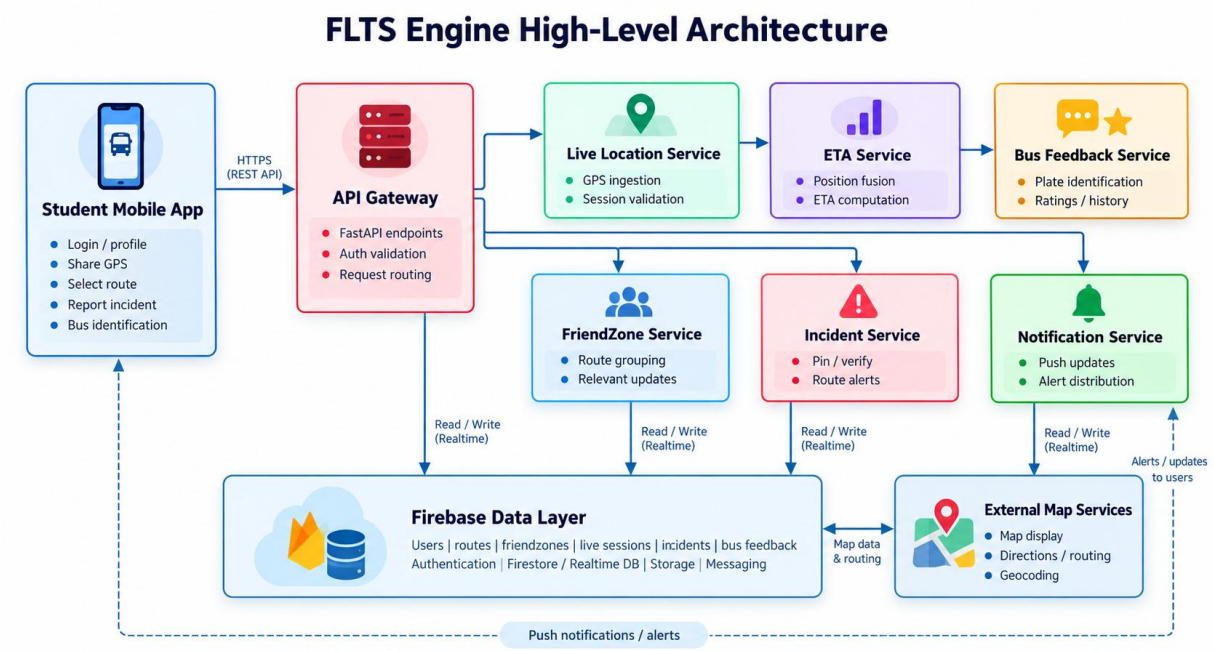


Figure 4.1 FLTS Engine high-level architecture

4.3 Design alternatives and decisions

A central design decision is to use passenger smartphones as the primary source of live data rather than requiring a tracker on each bus. This reduces dependence on dedicated infrastructure, but it also introduces challenges related to observation quality, participation, and privacy. The design addresses these concerns through session-based sharing, route validation, freshness checks, and, where appropriate, reliability weighting when observations are combined.

For ETA, the first implementation should favor an interpretable movement-based approach using current route progress and recent speed. More advanced machine-learning methods may be introduced later once sufficient historical trip data are available. This staged approach avoids presenting a complex prediction model before a suitable dataset exists.

For bus identification, manual plate entry is retained as a fallback even if image-based recognition is available. This supports usability when camera quality, lighting, or plate visibility reduces OCR accuracy.

4.4 Detailed design

4.4.1 Route-based FriendZones

A FriendZone is a dynamic, route-oriented community rather than a permanent social group. Users are associated according to the route identifier, travel direction, and an active travel or information session. A passenger travelling on a bus can contribute live data, while a student waiting along the same route can receive the resulting updates. This arrangement limits irrelevant information and defines a clear audience for ETA and incident updates.

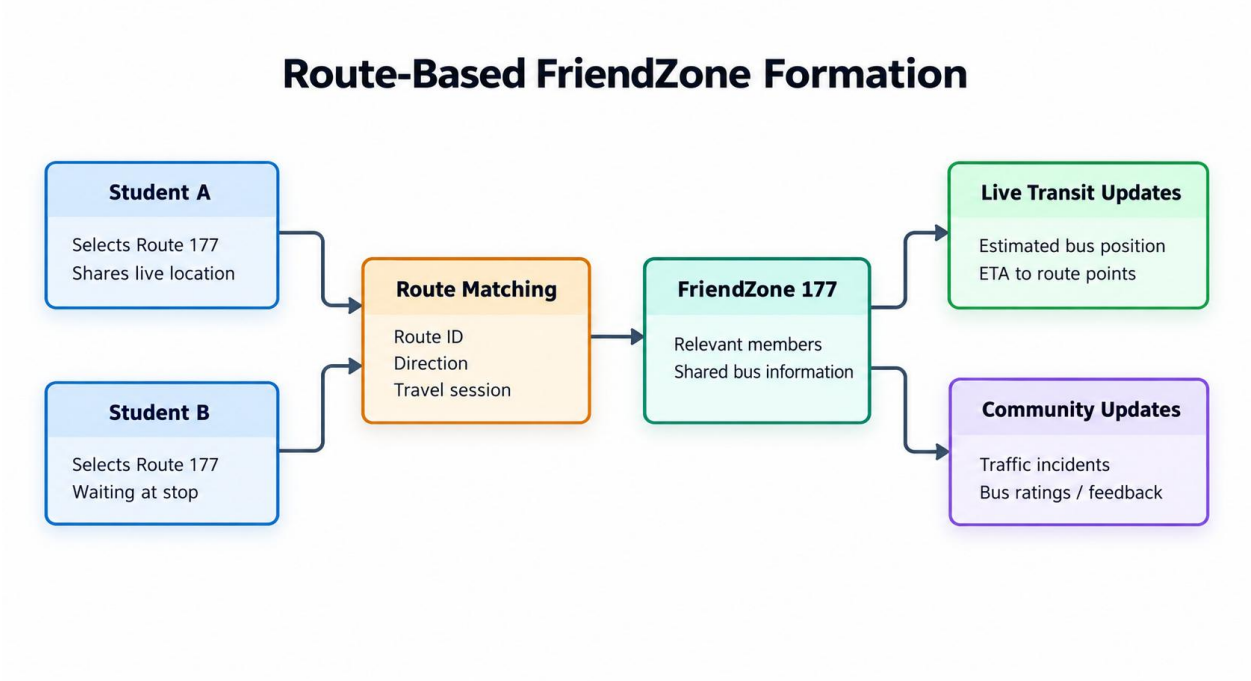


Figure 4.2 Route-based FriendZone formation

4.4.2 Collaborative bus position and ETA

Each observation includes latitude, longitude, timestamp, and estimated speed. When multiple users on the same bus or route contribute observations, the engine can calculate a weighted combined position, where more recent and reliable observations receive greater weight. The ShareMate research describes the conceptual bus position as a weighted combination of passenger positions. The resulting location is map-matched to the selected route and used to determine route progress and remaining travel distance.

ETA can initially be derived from remaining route distance and a smoothed estimate of recent speed, with adjustments for active incidents or persistent slow movement. The output should include the predicted arrival time, the last update time, and, where possible, a confidence indicator so users can distinguish a strong estimate from one based on sparse data.

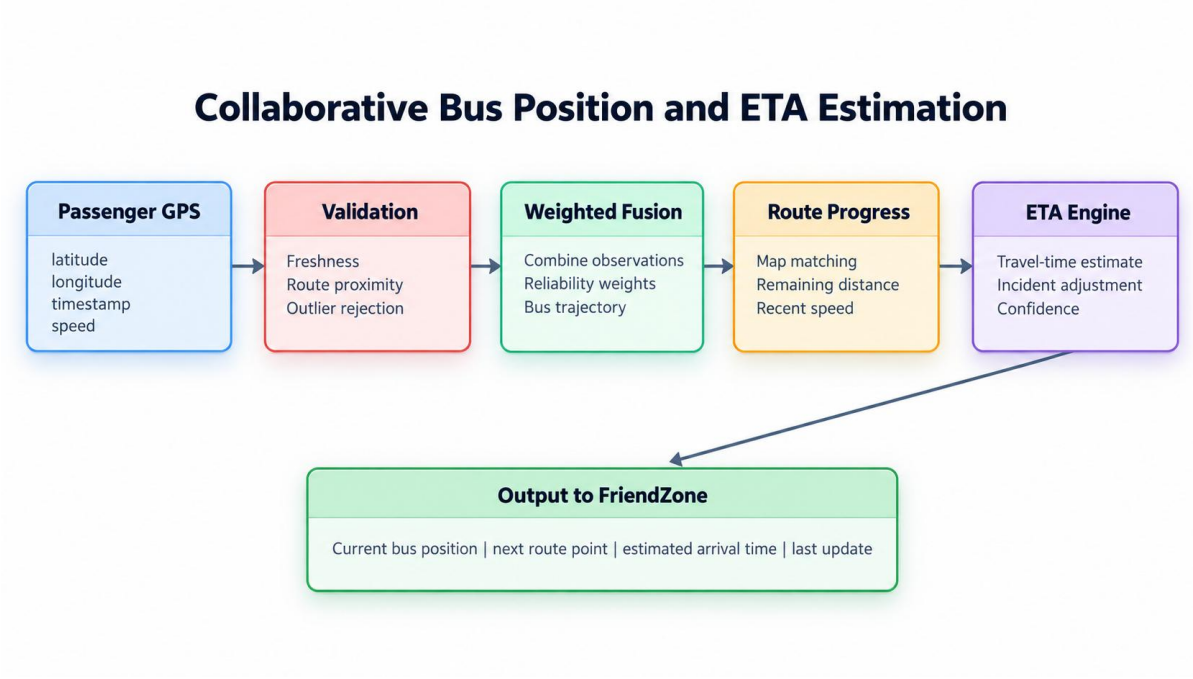


Figure 4.3 Collaborative bus position and ETA estimation

4.4.3 Traffic and incident reporting

The incident workflow allows a student to select an incident category, use the current location or place a pin on the map, optionally add a short description, and submit the report. The backend then associates the report with a nearby route and travel direction, applies validation rules, stores the record, and distributes the update to the relevant FriendZone. Duplicate or stale reports should be merged or expired to keep the map useful and current.



Figure 4.4 Traffic and incident reporting workflow

4.4.4 Bus identification and passenger feedback

The bus identification workflow uses the licence plate number as the key for retrieving a bus record. The number may be entered manually or extracted from an image using a recognition service. Once the plate has been validated, the system retrieves previous ratings and travel experiences. A student who has completed a journey can then submit new feedback, adding to the community service history associated with that bus.

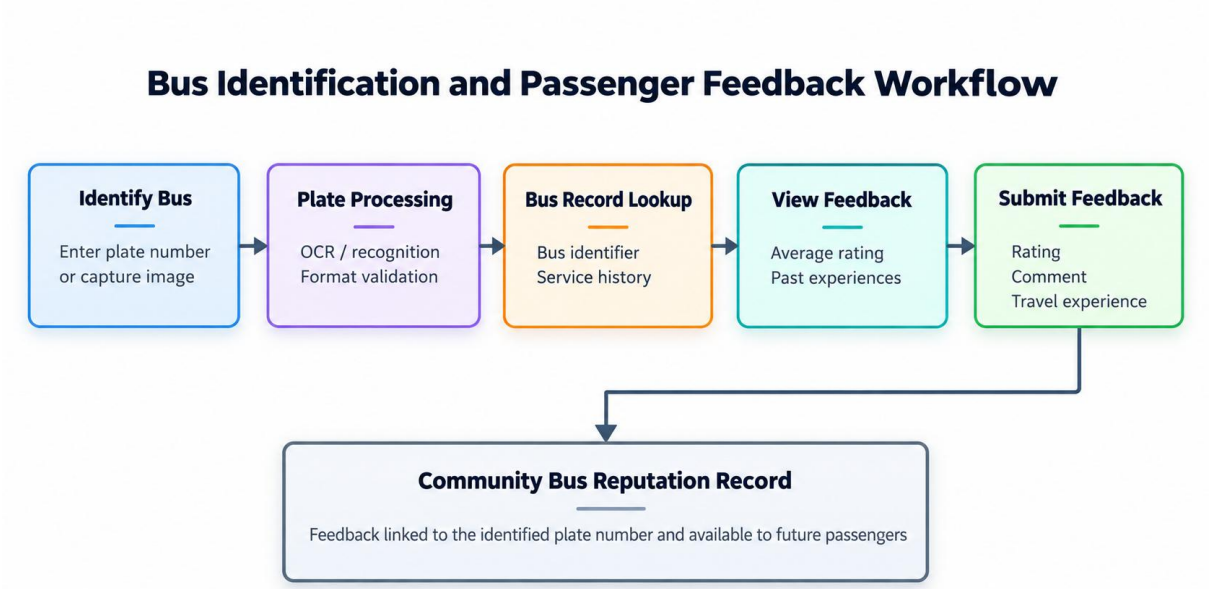


Figure 4.5 Bus identification and passenger feedback workflow

4.5 Standards and professional practice

The component should follow professional software-engineering practices including modular API design, HTTPS communication, least-privilege access control, environment-based secrets management, source control, code review, logging, structured testing, and privacy-aware data retention. Accessibility considerations should be applied to map labels, contrast, touch targets, error messages, and alternative text so that core functions are usable by a broad student population.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5.1 Development technology stack

Layer	Technology	Role
Mobile frontend	Flutter	Cross-platform mobile UI for Android/iOS.
Backend services	Python + FastAPI	REST APIs and transport-processing services.
Authentication / data	Firebase	Authentication, real-time data, storage, messaging.
Mapping	Cloud mapping API / Google Maps-compatible integration	Map display, route visualisation, geocoding and directions.
IDE	Visual Studio Code	Development, debugging, extensions and terminal workflow.
Version control	Git / GitHub	Source history, branches, collaboration and issue tracking.
API testing	Postman / Swagger UI	Endpoint verification during development.
Local service exposure	Ngrok during development	Temporary secure service links for integration testing.

The ShareMate research paper specifies Flutter for the frontend, Python for backend services, Firebase for data management and real-time communication, a cloud mapping API, and Ngrok for secure development-time links. The implementation plan in this thesis keeps those architectural roles and organizes the FLTS logic into focused services.

5.2 Component implementation

Authentication and profiles

Firebase Authentication manages sign-in and user identity. Additional university and route preferences are stored in the user profile. Backend endpoints validate authenticated requests before allowing access to protected FLTS operations.

Live location sharing

The mobile client requests location permission only when required. During an active sharing session, timestamped observations are sent to the live-location service and stored with route/session identifiers. Sharing stops explicitly when the user ends the session.

FriendZone service

Route and direction information are used to determine the FriendZone. The service exposes active bus sessions, incident information, and ETA outputs only for relevant route users.

ETA service

The service retrieves recent location observations, filters invalid or stale data, computes an estimated bus position and route progress, and returns ETA for configured route points.

Incident service

Incident reports are persisted with category, coordinates, route association, reporter identifier, and timestamps. Expiration and moderation rules keep active map information current.

Bus feedback service

A validated plate number is used to retrieve the bus record. Feedback is stored separately from user profile data and aggregated for presentation as ratings and travel experiences.

5.3 Integration with the other components

FLTS shares the ShareMate identity and mobile application shell with the other project components. The component should not directly depend on internal implementation details of ANIE, FriendZone Ride, or Grocery services. Integration occurs through common user identity, shared notification infrastructure, API contracts, and navigation in the mobile client. This preserves loose coupling and allows each research component to be deployed or tested independently.

5.4 Verification and software testing

Table 5.2 FLTS verification test cases

Test ID	Test	Input / precondition	Expected result
TC-FLTS-01	Register/login with valid university user	Valid credentials	User authenticated and profile loaded.
TC-FLTS-02	Start live bus sharing	Permission granted, route selected	Session created and GPS updates accepted.
TC-FLTS-03	Reject stale location	Old timestamp	Observation ignored or flagged as stale.
TC-FLTS-04	FriendZone grouping	Two users on same route	Both receive the same route-relevant updates.
TC-FLTS-05	ETA calculation	Valid recent observations	ETA returned for selected route point.
TC-FLTS-06	Submit incident pin	Accident + valid map location	Incident stored and visible to same-route users.
TC-FLTS-07	Identify bus by plate	Valid plate	Matching bus record and feedback displayed.
TC-FLTS-08	Submit passenger feedback	Authenticated user + bus record	Rating/comment stored and aggregate updated.
TC-FLTS-09	Unauthorized write attempt	Missing/invalid token	Request rejected.
TC-FLTS-10	Stop sharing	Active session	Location updates cease and session closes.

5.5 Deployment and operation

The final deployment should separate development, test, and production environments. API secrets, service-account keys, and map credentials must be stored outside source control. Logging should capture service health and errors without recording unnecessary precise location or personal information. Database security rules, API authentication, request validation, backups, monitoring, and rate limits should be configured before a pilot with real students.

6 RESULTS AND VALIDATION

6.1 Research results

The ShareMate research paper reports preliminary quantitative results for the FLTS component. ETA prediction accuracy is reported as 91.4%, while GPS/location processing accuracy is reported as 94.2%. These figures provide initial evidence that the passenger-generated approach can support useful transport information. Because the research paper does not provide the full FLTS sample size, raw confusion/error distribution, or detailed route-by-route dataset in the supplied material, this thesis treats the values as preliminary reported results rather than reproducing an independent statistical analysis.

Table 6.1 Preliminary FLTS results reported in ShareMate research

Metric	Reported result	Interpretation
ETA prediction accuracy	91.4%	Reported for FLTS Engine in ShareMate evaluation.
GPS/location processing accuracy	94.2%	Reported for FLTS Engine in ShareMate evaluation.
Overall ShareMate average response time	420 ms	System-level result, not FLTS-only.
Full ShareMate user satisfaction	4.5/5	System-level comparative result, not FLTS-only.

6.2 Product and technical results

At the product level, the design demonstrates how a passenger can become the data source for live public-transport information. The architecture supports a complete path from mobile GPS capture through route grouping and ETA processing to live map output. Incident reporting and bus feedback extend the component beyond tracking by introducing community context and service reputation.

6.3 User or stakeholder validation

The supplied ShareMate paper reports a user-satisfaction value of 4.5/5 for the full integrated ShareMate system; this result cannot be attributed solely to FLTS. A dedicated FLTS validation should therefore be conducted with university commuters using tasks such as locating an active bus, interpreting an ETA, reporting an incident, identifying a bus, and reviewing or submitting feedback. Task completion, perceived usefulness, trust, privacy concerns, and overall satisfaction should be recorded separately for the individual component.

6.4 Integration results

The overall ShareMate evaluation reports an average response time of 420 ms and shows that the integrated system can combine location, trust, and user-specific information. For FLTS, successful integration means that authenticated student identity, route data, live GPS updates, incident records, and feedback can move between the mobile client and backend services without requiring direct coupling to other ShareMate engines.

6.5 Summary of findings

The preliminary evidence supports the technical feasibility of the FLTS concept. The reported results indicate that a passenger-powered approach can process location data with high accuracy while producing useful ETA outputs. Stronger validation, however, requires transparent route-level datasets, repeatable experiments, broader student participation, and explicit measurement of privacy, battery use, network reliability, and plate-recognition performance.

7 DISCUSSION

7.1 Interpretation of findings

The reported FLTS results suggest that crowdsourced passenger data can serve as a practical source of live transport information. The reported 94.2% GPS/location processing accuracy indicates that the data-processing pipeline handled passenger observations reliably under the stated evaluation, while the 91.4% ETA accuracy suggests that movement information can be translated into useful arrival estimates. The lower ETA performance is reasonable because ETA also depends on future traffic conditions, stop delays, and changes in vehicle speed.

7.2 Comparison with literature baselines and existing solutions

The FLTS approach is consistent with earlier work showing that real-time and historical transport data can support bus arrival prediction [7]–[9], [19] and that crowdsourced incident reports can add timely traffic information [10]. Its distinction is architectural and contextual: the component combines these ideas around a route-based university community and treats passenger phones as the sensing infrastructure. The bus identification and feedback mechanism further incorporates reputation concepts [4]–[6] and vehicle-recognition research [12].

7.3 Research contribution

The research contribution is a conceptual and evaluable framework for decentralized, student-centered public-transport intelligence. It brings passenger observations, route-based communities, collaborative prediction, incident reporting, and vehicle reputation together as one research component. The framework also makes explicit the methodological requirements of this approach, including observation reliability, privacy, trust, and route relevance.

7.4 Product contribution

The product contribution is the FLTS Engine within ShareMate. It provides the wider platform with real-time public-transport functionality that complements private ride sharing and other student community services. FLTS can operate as a standalone

transport module while still using shared authentication, notifications, and mobile navigation within ShareMate.

7.5 Limitations and threats to validity

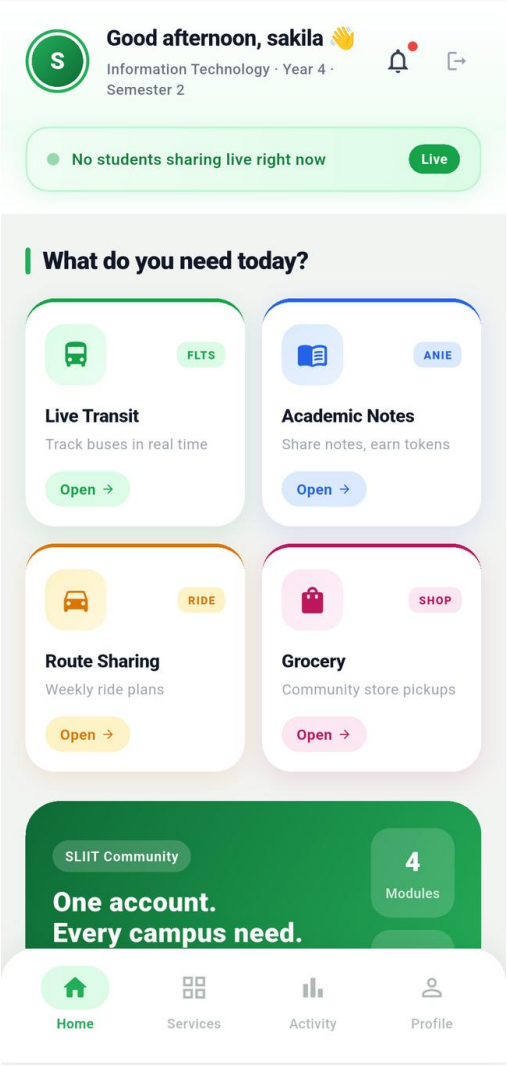
Table 7.1 Threats to validity and mitigation

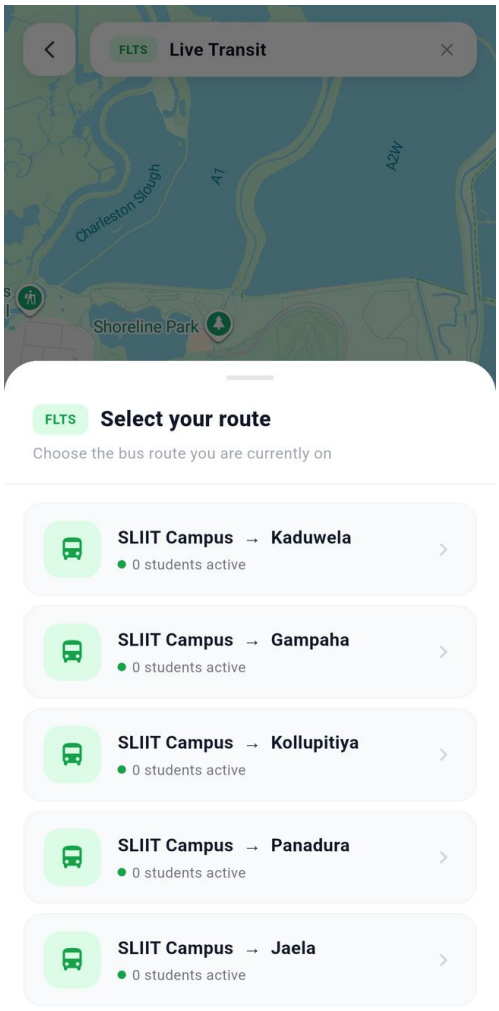
Threat	Effect	Mitigation
Participation bias	Trips with few participating passengers may have weaker live coverage.	Pilot on repeated routes; show freshness/confidence indicators.
GPS noise	Urban environments and phone hardware can introduce location error.	Route matching, outlier filtering, smoothing, multi-user fusion.
ETA variability	Traffic, dwell time, and unexpected events affect arrival estimates.	Use rolling speed, incident adjustment, and later ML models.
Incorrect incident reports	Community reports may be duplicated or false.	Confirmation, expiry, moderation, reporter reputation.
Plate recognition errors	Lighting, angle, and image quality reduce OCR accuracy.	Manual fallback and validation rules.
Privacy concerns	Location data can reveal sensitive movement patterns.	Explicit consent, minimization, access control, retention limits.
Evaluation completeness	Supplied paper lacks full FLTS raw evaluation dataset.	Future reproducible route-level evaluation with transparent data.

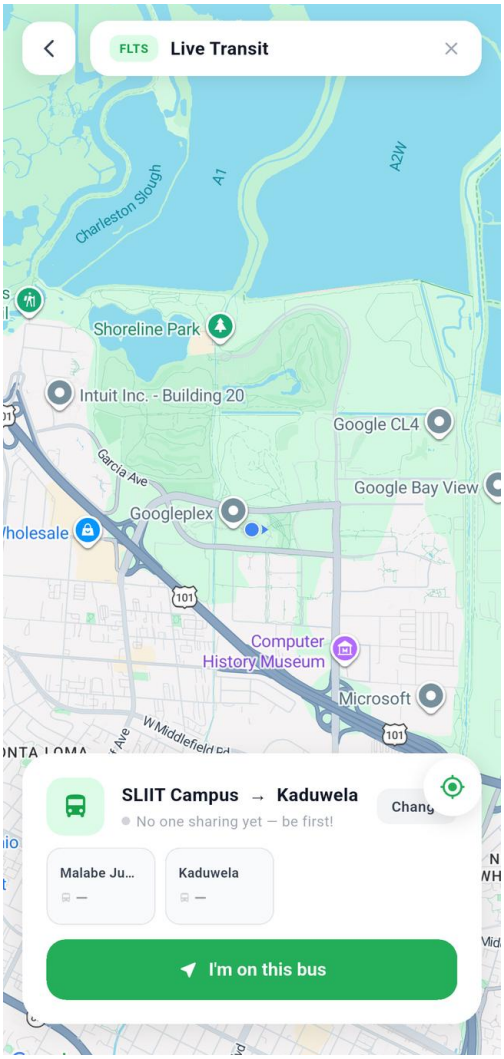
7.6 Implications

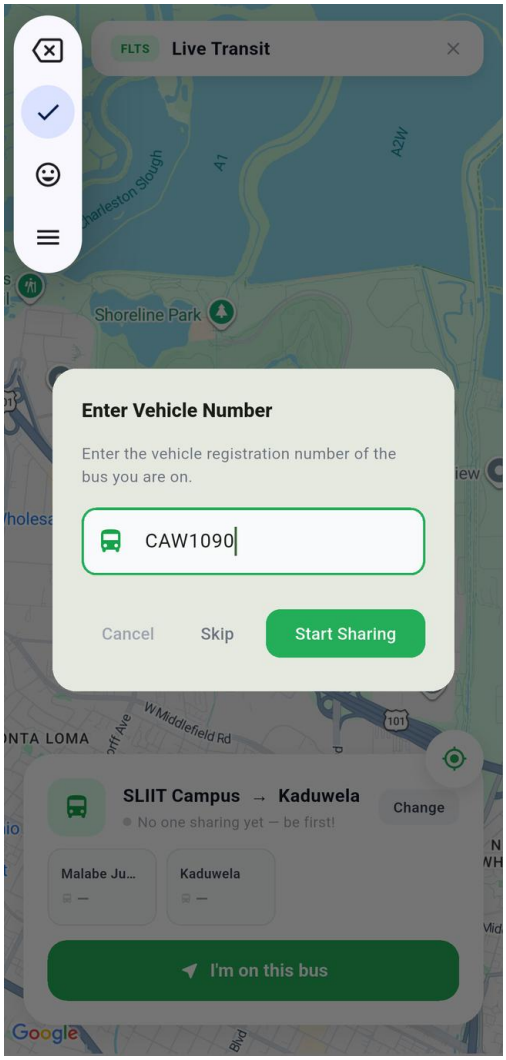
If validated at larger scale, the FLTS model could improve transport visibility in environments where official real-time feeds are unavailable. The approach may be particularly suitable for university commuting corridors because repeated routes and community identity can support participation and trust. The same design principles could later be generalized beyond universities, but broader public deployment would require stronger moderation, fraud prevention, accessibility, and operational partnerships.

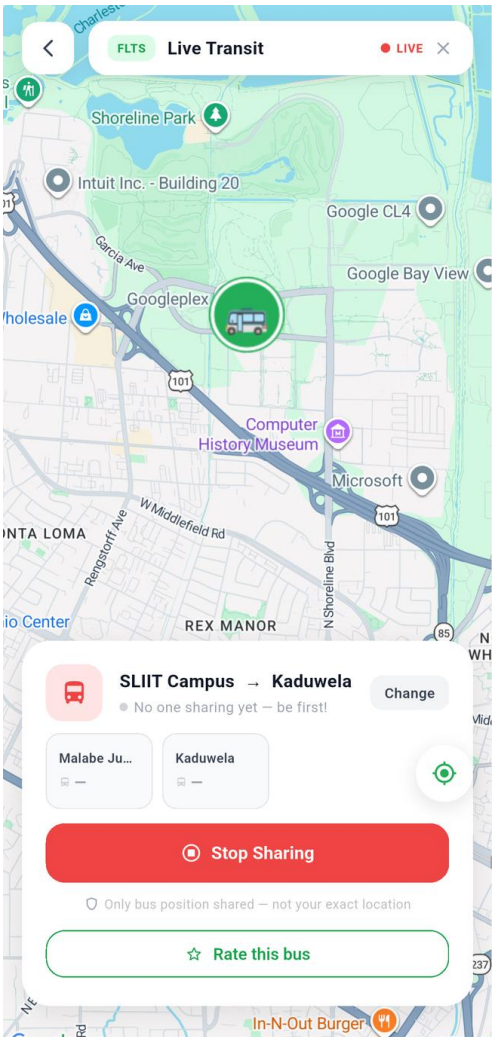
7.7 User interface design

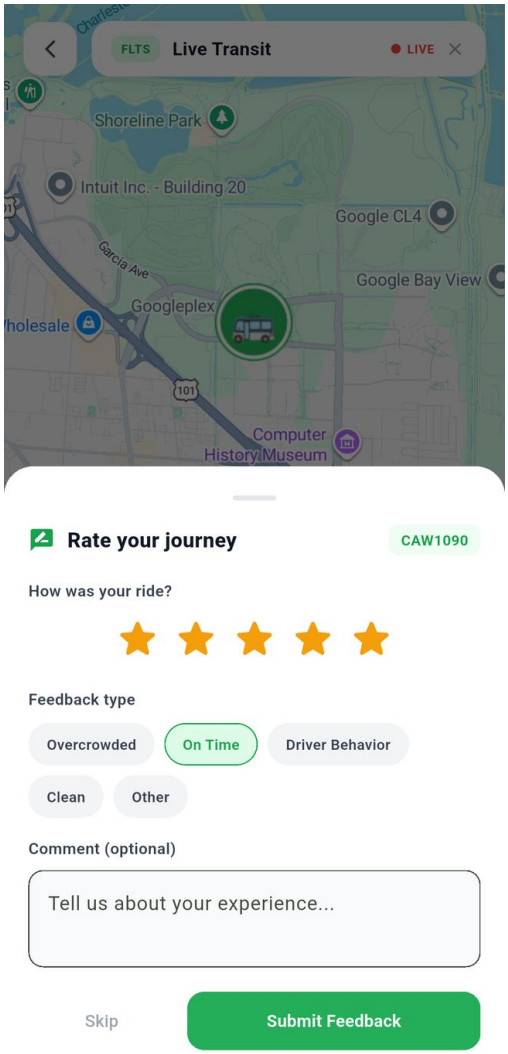





FLTS
Live Transit
Charleston Slough
A1
A2W
Shoreline Park
Intuit Inc. - Building 20
Google CL4
Garcia Ave
Google Bay View
Googleplex
Computer History Museum
Microsoft
101
W Middlefield Rd
SLIIT Campus → Kaduwela
No one sharing yet — be first!
Malabe Ju...
Kaduwela
I'm on this bus


FLTS
Live Transit
Shoreline Park
Enter Vehicle Number
Enter the vehicle registration number of the bus you are on.
CAW1090
Cancel
Skip
Start Sharing
W Middlefield Rd
101
SLIIT Campus → Kaduwela
No one sharing yet — be first!
Change
Malabe Ju...
Kaduwela
I'm on this bus
Google


FLTS
Live Transit
LIVE
Shoreline Park
Intuit Inc. - Building 20
Google CL4
Garcia Ave
Google Bay View
Googleplex
101
Computer History Museum
Microsoft
W Middlefield Rd
Rengstorff Ave
N Shoreline Blvd
REX MANOR
SLIIT Campus → Kaduwela
No one sharing yet — be first!
Change
Malabe Ju...
Kaduwela
Stop Sharing
Only bus position shared — not your exact location
Rate this bus
In-N-Out Burger


FLTS
Live Transit
LIVE
Shoreline Park
Intuit Inc. - Building 20
Google CL4
Garcia Ave
Google Bay View
Googleplex
101
Computer History Museum
Rate your journey
CAW1090
How was your ride?
Feedback type
Overcrowded
On Time
Driver Behavior
Clean
Other
Comment (optional)
Tell us about your experience...
Skip
Submit Feedback

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The primary users are university students who rely on public buses and need better real-time travel awareness. The value proposition is low-cost access to live bus movement, ETA, route incidents, and bus service history without requiring a tracker on every bus. Secondary stakeholders include universities, student unions, transport operators, local businesses, and mobility partners interested in better understanding student commuting.

8.2 Market and competitive position

FLTS is positioned between official transport tracking systems and community traffic applications. Its competitive differentiation is the university-centered FriendZone model and the integration of passenger-powered bus tracking with ETA, incidents, and bus reputation. The strategy should avoid claiming to replace official systems; instead, FLTS can complement them where official feeds are incomplete and integrate official data later where available.

8.3 Business or sustainability model

A sustainable model should keep core safety and basic transport visibility accessible to students. Potential revenue options include university or institutional subscriptions for analytics and deployment support, premium convenience features, carefully controlled campus advertising, and partnerships with transport or mobility organizations. Any monetization of location or personal data should be avoided; user trust is central to participation.

8.4 Scalability intellectual property and compliance

The microservice architecture supports horizontal scaling of high-demand services such as location ingestion and notifications. Intellectual property may include the ShareMate brand, software code, UI assets, documentation, and original service logic, subject to institutional ownership rules. Compliance priorities include

data-protection obligations, consent management, secure cloud processing, third-party API terms, and clear policies for user-generated incidents and feedback.

8.5 Contribution to the Sustainable Development Goals

Table 8.1 Commercialisation and sustainability considerations

SDG	Contribution
SDG 9 – Industry, Innovation and Infrastructure	Uses digital infrastructure and mobile crowdsensing to improve access to transport information.
SDG 11 – Sustainable Cities and Communities	Supports more informed public-transport use and community awareness of route conditions.
SDG 12 – Responsible Consumption and Production	Encourages shared use of existing transport and reduces unnecessary travel uncertainty within the broader ShareMate ecosystem.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the FriendZone LiveTransit Share Engine, an individual ShareMate component designed to improve public-transport visibility for university students through community-generated data. FLTS replaces the assumption of mandatory bus-installed tracking hardware with a passenger-powered model in which authenticated students contribute GPS observations during travel. The resulting information is organized through route-based FriendZones and converted into live bus position, ETA, and route-specific incident awareness. Bus identification and feedback add a service-reputation layer that helps future passengers understand previous experiences associated with a particular vehicle.

The preliminary results reported in the ShareMate research paper provide evidence of technical feasibility, including 94.2% GPS/location processing accuracy and 91.4% ETA prediction accuracy. These results are promising, but the individual component still requires transparent large-scale field validation to establish accuracy, robustness, privacy, and usability under varied routes and participation levels.

9.2 Achievement of objectives

The design addresses all six stated objectives at the architecture and research-planning level: authenticated GPS sharing, FriendZone formation, collaborative position/ETA estimation, incident pinning, bus identification with feedback, and component evaluation. The final implementation should be assessed against the success indicators in Table 1.1 before claiming complete objective achievement.

9.3 Individual research and product contributions

The individual research contribution is the integration of crowdsourced sensing, route communities, prediction, incident reporting, and reputation into a unified public-transport framework for university students. The product contribution is the FLTS microservice domain within ShareMate, including the data model, API boundaries, workflows, and mobile interactions required for real-time public transport sharing.

9.4 Future directions

- Train and compare advanced ETA models once a sufficiently large trip-history dataset is available.
- Introduce observation reliability scores and fraud/anomaly detection for suspicious location contributions.
- Evaluate battery and mobile-data consumption under different GPS update intervals.
- Improve incident trust using confirmations, expiry logic, and route-based reputation.
- Evaluate licence plate recognition on Sri Lankan bus plates under varied lighting and camera conditions.
- Pilot the system with multiple universities and routes, then study expansion to the wider public.
- Integrate official transport feeds where available so crowdsourced and authority data can complement each other.

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GLOSSARY

Term	Definition
Collaborative ETA	An arrival-time estimate derived from shared passenger observations rather than a single dedicated vehicle tracker.
FriendZone	A dynamic route-based community of students who share or receive transit information relevant to the same route.
Live bus session	A time-bounded period during which one or more passengers contribute observations associated with a bus journey.
Mobile crowdsensing	Collection of sensor observations from participating mobile devices to infer a shared real-world condition.
Passenger-powered tracking	The FLTS approach of using passenger smartphones as the source of bus movement data.
Route matching	The process of associating a user, observation, or incident with the appropriate bus route and direction.

APPENDICES

Appendix A Research instruments or extended results

A.1 Suggested FLTS student validation questionnaire

1. How often do you travel to university using public buses?
2. How difficult is it to know when your bus will arrive?
3. How useful would live bus location sharing from other passengers be?
4. How easy was it to join the correct FriendZone?
5. How understandable was the ETA shown by the application?
6. How useful were traffic/incident pins on the route map?
7. How useful was the bus feedback history?
8. How comfortable are you with sharing GPS location only during an active bus trip?
9. How much do you trust community-generated transport information?
10. Overall, how satisfied are you with the FLTS component?

Suggested scale for questions 3–10: 1 = Strongly disagree / Very poor, 5 = Strongly agree / Excellent.

A.2 Extended test evidence to include in the final submission

- Screenshots of registration/login and permission flow.
- Example live bus sharing session with timestamps.
- FriendZone route membership evidence.
- ETA prediction versus observed arrival-time table.
- Incident pin creation and route delivery screenshots.
- Bus plate identification and feedback history screenshots.
- API test outputs and selected logs with personal data removed.
- UAT summary and anonymized participant feedback.

Appendix B Academic integrity and AI use disclosure

Draft disclosure to be adjusted to the module's official policy before submission: Generative AI tools were used to assist with outlining, language refinement,

document structuring, and drafting selected explanatory text. The student remains responsible for verifying all technical content, source citations, calculations, implementation details, experimental results, and conclusions. No AI-generated result is presented as an independently measured experimental result unless it was verified against the project's actual evidence. Source code, datasets, and final analyses used for assessment must be reviewed and understood by the student.

Representative workflow: (1) provide the official thesis template and project research paper; (2) request a structured draft aligned to the individual FLTS component; (3) verify all claims against the supplied research material; (4) revise architecture, implementation, and results sections to match the actual system; and (5) update the disclosure to comply with the latest module instructions.